

B2 Symmetry and Relativity — Model Solutions, 2016

Question 1 — Kinematics, hyperbolic motion, constant force

Problem. Define τ , U , A and find A in terms of γ , $\mathbf{v}$, $\mathbf{a}$. Find invariants of U and P and evaluate $U \cdot A$. Discuss time-like intervals and meaning of $D \cdot B = 0$. Hyperbolic motion $x^2 - t^2 = L^2$: find v , γ , a and show $a\gamma^3$ constant. Constant 3-force $\mathbf{f}$ from rest: find $\beta(t), \gamma(t)$, sketch. Numerical: electron in $E = 5 \text{ MV/m}$ over $L = 10 \text{ m}$.

(a) 4-velocity, 4-acceleration and invariants

Proper time:

$$d\tau = dt/\gamma.$$

4-velocity:

$$U^\mu = \frac{dX^\mu}{d\tau} = \gamma(c, \mathbf{v}).$$

4-acceleration:

$$A^\mu = \frac{dU^\mu}{d\tau} = \gamma \frac{d}{dt} [\gamma(c, \mathbf{v})].$$

Use $d\gamma/dt = \gamma^3(\mathbf{v} \cdot \mathbf{a})/c^2$:

$$A^\mu = \gamma \left(\gamma^3 \frac{\mathbf{v} \cdot \mathbf{a}}{c}, \gamma^3 \frac{\mathbf{v} \cdot \mathbf{a}}{c^2} \mathbf{v} + \gamma^2 \mathbf{a} \right).$$

Boxed:

$$A^\mu = \left(\gamma^4 \frac{\mathbf{v} \cdot \mathbf{a}}{c}, \gamma^4 \frac{(\mathbf{v} \cdot \mathbf{a}) \mathbf{v}}{c^2} + \gamma^2 \mathbf{a} \right)$$

Invariants (Minkowski signature $(+, -, -, -)$):

$$U \cdot U = \gamma^2(c^2 - v^2) = c^2, \quad P \cdot P = m^2 c^2.$$

Differentiate $U \cdot U = c^2$ w.r.t. τ :

$$U \cdot A = 0$$

(b) Time-like interval and $D \cdot B = 0$

Separation $\Delta = D - B = (c\Delta t, \Delta \mathbf{x})$ time-like:

$$c^2(t_d - t_b)^2 - |\mathbf{x}_d - \mathbf{x}_b|^2 > 0$$

Time-like means a signal at speed $\leq c$ connects them (causal). Order is invariant: *no* frame exists in which they are simultaneous (that would require a space-like separation).

$D \cdot B = c^2 t_d t_b - \mathbf{x}_d \cdot \mathbf{x}_b = 0$ is a frame-invariant orthogonality of the two 4-position vectors. Physically: in the frame where one event occurs at the spatial origin (say $\mathbf{x}_b = 0$ at $t = t_b$), $D \cdot B = c^2 t_d t_b$; orthogonality forces $t_d t_b = 0$, i.e. one event lies on the light-cone-like hypersurface $t = 0$ of the other.

(c) Hyperbolic motion

Proper acceleration: magnitude of A^μ in the instantaneous rest frame, $\alpha = \sqrt{-A \cdot A}$.

Pure force: 3-force whose action leaves rest mass invariant, equivalently $F^\mu = dP^\mu/d\tau$ with $F \cdot U = 0$.

Worldline ($c = 1$) $x^2 - t^2 = L^2$. Differentiate w.r.t. t :

$$2x\dot{x} - 2t = 0.$$

Read off:

$$v = \dot{x} = t/x.$$

Lorentz factor:

$$\gamma = (1 - v^2)^{-1/2} = (1 - t^2/x^2)^{-1/2} = x/L.$$

Differentiate $v = t/x$:

$$a = \dot{v} = \frac{1}{x} - \frac{t\dot{x}}{x^2} = \frac{1}{x} - \frac{t^2}{x^3} = \frac{x^2 - t^2}{x^3} = \frac{L^2}{x^3}.$$

Hence:

$$v = t/x, \quad \gamma = x/L, \quad a = L^2/x^3$$

Combine:

$$a\gamma^3 = \frac{L^2}{x^3} \cdot \frac{x^3}{L^3} = \frac{1}{L}.$$

$$a\gamma^3 = 1/L = \text{const}$$

(d) Constant 3-force from rest

$\mathbf{f} = d\mathbf{p}/dt = \text{const}$, $\mathbf{p}(0) = 0$. Integrate:

$$\mathbf{p}(t) = \mathbf{f}t.$$

$p = \gamma mv$ and $E^2 = (pc)^2 + (mc^2)^2$:

$$E(t) = \sqrt{(fct)^2 + (mc^2)^2}.$$

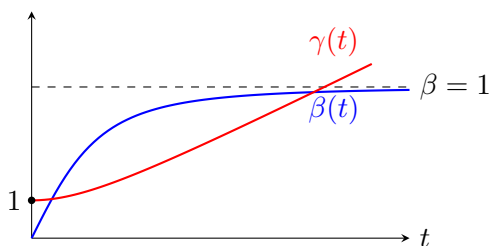
$v = pc^2/E$ gives

$$\beta(t) = \frac{ft/(mc)}{\sqrt{1 + (ft/mc)^2}}$$

$\gamma = E/(mc^2)$:

$$\gamma(t) = \sqrt{1 + (ft/mc)^2}$$

Limits: $\beta \rightarrow ft/(mc)$ for small t (Newtonian); $\beta \rightarrow 1$, $\gamma \rightarrow ft/(mc)$ for large t .



(e) **Numerical: electron through $L = 10$ m at $E = 5$ MV/m**

Total voltage:

$$V = EL = 5 \times 10^7 \text{ V}.$$

Use the hint $\gamma = 1 + eV/(m_0c^2)$ with $m_0c^2 = 0.511$ MeV:

$$\gamma = 1 + \frac{50}{0.511} = 1 + 97.85 = 98.85.$$

$$\boxed{\gamma \approx 98.9}$$

$$\beta = \sqrt{1 - 1/\gamma^2} = \sqrt{1 - 1.024 \times 10^{-4}}.$$

Binomial:

$$\beta \approx 1 - \frac{1}{2\gamma^2} = 1 - 5.12 \times 10^{-5}.$$

$$\boxed{\beta \approx 0.99995}$$

20% reduction: $V \rightarrow 0.8V = 40$ MV, $\gamma' = 1 + 40/0.511 = 79.28$:

$$1 - \beta' \approx \frac{1}{2\gamma'^2} = 7.95 \times 10^{-5}.$$

$$\Delta\beta = \beta - \beta' \approx (7.95 - 5.12) \times 10^{-5} \approx 2.8 \times 10^{-5}.$$

$$\boxed{\Delta\beta \approx 3 \times 10^{-5}}$$

Time of flight: with constant 3-force $f = eE$,

$$x(t) = \frac{mc^2}{f} \left(\sqrt{1 + (ft/mc)^2} - 1 \right) = \frac{mc^2}{eE} (\gamma - 1).$$

Solve for t :

$$t = \frac{mc}{eE} \sqrt{\gamma^2 - 1} = \frac{mc\gamma\beta}{eE}.$$

Numerically, $mc^2/e = 0.511 \times 10^6$ V, $E = 5 \times 10^6$ V/m, $c = 3 \times 10^8$ m/s:

$$t = \frac{(0.511 \times 10^6 \text{ V}) \gamma \beta}{(5 \times 10^6 \text{ V/m}) \cdot c} = \frac{0.511}{5 \cdot 3 \times 10^8} \cdot 98.85 \cdot 0.99995 \text{ s}.$$

$$t = \frac{0.511 \cdot 98.84}{1.5 \times 10^9} \text{ s} = \frac{50.5}{1.5 \times 10^9} \text{ s}.$$

$$\boxed{t \approx 3.4 \times 10^{-8} \text{ s}}$$

Question 2 — Velocity transformation, helical motion, photon kinematics

(a) **Aberration formula**

S' moves with $\mathbf{v} = (v, 0, 0)$ relative to S . Velocity transformation:

$$u'_{\parallel} = \frac{u_{\parallel} - v}{1 - u_{\parallel}v/c^2}, \quad u'_{\perp} = \frac{u_{\perp}}{\gamma_v(1 - u_{\parallel}v/c^2)}.$$

With $u_{\parallel} = u \cos \theta$, $u_{\perp} = u \sin \theta$:

$$\tan \theta' = \frac{u'_{\perp}}{u'_{\parallel}} = \frac{u \sin \theta / [\gamma_v (1 - u \cos \theta v / c^2)]}{(u \cos \theta - v) / (1 - u \cos \theta v / c^2)}.$$

Cancel the common denominator:

$$\boxed{\tan \theta' = \frac{u \sin \theta}{\gamma_v (u \cos \theta - v)}}$$

(b) Decelerated helical electron

Initial state: $\gamma_0 = 17$, $p_{\parallel} = p_{\perp}$, so $p_{\parallel}^2 = p_{\perp}^2 = p^2/2$. Energy-momentum: $E_0^2 = (p_0 c)^2 + (mc^2)^2$, $E_0 = \gamma_0 mc^2$:

$$p_0 c = mc^2 \sqrt{\gamma_0^2 - 1}.$$

Longitudinal piece:

$$p_{\parallel,0} c = p_0 c / \sqrt{2} = (mc^2 / \sqrt{2}) \sqrt{\gamma_0^2 - 1}.$$

E_z does no work on transverse motion: $p_{\perp}$ is conserved. Hence

$$p_{\perp,f} = p_{\perp,0} = p_0 / \sqrt{2}.$$

At the end $v_z = 0$, so $p_{\parallel,f} = 0$ and the kinetic energy lost equals work done:

$$E_0 - E_f = e E_z \cdot 1 \text{ m}.$$

Work-energy with $V = E_z \cdot 1 \text{ m}$ and the hint $\gamma = 1 + eV/(mc^2)$ but applied to the longitudinal part: cleaner is energy conservation directly. After the field, $p_{\parallel,f} = 0$ so

$$E_f^2 = (p_{\perp,0} c)^2 + (mc^2)^2 = \frac{1}{2} (p_0 c)^2 + (mc^2)^2.$$

Substitute $(p_0 c)^2 = (mc^2)^2 (\gamma_0^2 - 1)$:

$$E_f^2 = (mc^2)^2 \left[\frac{1}{2} (\gamma_0^2 - 1) + 1 \right] = (mc^2)^2 \frac{\gamma_0^2 + 1}{2}.$$

Hence

$$\gamma_f = \sqrt{(\gamma_0^2 + 1)/2}.$$

With $\gamma_0 = 17$: $\gamma_f = \sqrt{(289 + 1)/2} = \sqrt{145} = 12.04$.

$$\boxed{\gamma_f \approx 12.0}$$

Voltage:

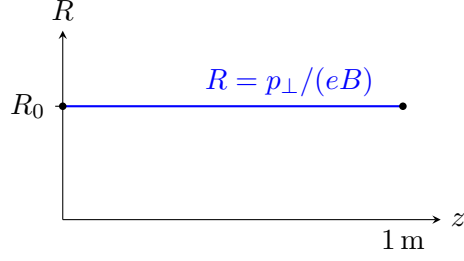
$$V = (\gamma_0 - \gamma_f) mc^2 / e = (17 - 12.04) \cdot 0.511 \text{ MV} = 2.535 \text{ MV}.$$

Field strength over 1 m:

$$\boxed{E_z \approx 2.5 \text{ MV/m}}$$

Larmor radius along z : $R(z) = p_{\perp}(z)/(eB)$. Since $p_{\perp}$ is unchanged by $E_z \parallel \hat{z}$,

$$\boxed{R(z) = \text{const}}$$



(c) Two-photon system

Both photons have the same ω . 4-momenta:

$$P_1 = \frac{\hbar\omega}{c}(1, 1, 0, 0), \quad P_2 = \frac{\hbar\omega}{c}(1, \cos\theta, \sin\theta, 0).$$

Total:

$$P = P_1 + P_2 = \frac{\hbar\omega}{c}(2, 1 + \cos\theta, \sin\theta, 0).$$

Invariant mass squared:

$$P \cdot P = \left(\frac{\hbar\omega}{c}\right)^2 [4 - (1 + \cos\theta)^2 - \sin^2\theta].$$

Expand:

$$(1 + \cos\theta)^2 + \sin^2\theta = 1 + 2\cos\theta + \cos^2\theta + \sin^2\theta = 2 + 2\cos\theta.$$

Hence

$$P \cdot P = \left(\frac{\hbar\omega}{c}\right)^2 \cdot (2 - 2\cos\theta) = \left(\frac{\hbar\omega}{c}\right)^2 \cdot 4\sin^2(\theta/2).$$

Rest energy $E_* = c\sqrt{P \cdot P}$:

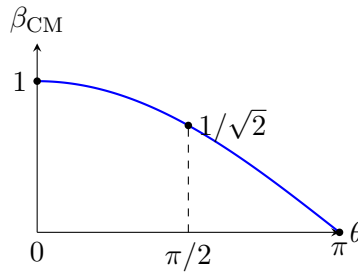
$$E_* = 2\hbar\omega |\sin(\theta/2)|$$

CM velocity: $\mathbf{v}_{\text{CM}} = c^2 \mathbf{P}_{\text{tot}} / E_{\text{tot}}$. With $E_{\text{tot}}/c = 2\hbar\omega/c$ and $|\mathbf{P}_{\text{tot}}| = (\hbar\omega/c)\sqrt{(1 + \cos\theta)^2 + \sin^2\theta}$:

$$|\mathbf{P}_{\text{tot}}|/(E_{\text{tot}}/c) = \frac{1}{2}\sqrt{2 + 2\cos\theta} = |\cos(\theta/2)|.$$

$$\beta_{\text{CM}} = |\cos(\theta/2)|$$

Direction: along the bisector of $\hat{P}_1$ and $\hat{P}_2$, angle $\theta/2$ above the x -axis. $\beta_{\text{CM}} = 1$ at $\theta = 0$ (photons collinear; no rest frame), $\beta_{\text{CM}} = 0$ at $\theta = \pi$ (head-on; CM at rest in S).



(d) Inverse Compton scattering

Maximum scattered photon energy occurs in head-on collision and back-scattering. Boost to electron rest frame, scatter, boost back. Doppler factor for photon coming toward electron (angle π to $\mathbf{v}$):

$$\omega^* = \gamma\omega(1 + \beta) \quad (\text{blueshift in electron frame}).$$

Compton formula in electron frame, back-scattering ($\theta^* = \pi$):

$$\frac{1}{\omega^{**}} - \frac{1}{\omega^*} = \frac{2\hbar}{m_e c^2}.$$

Boost back; back-scattered photon now moves with $+\mathbf{v}$, gets blueshifted again:

$$\omega' = \gamma\omega^{**}(1 + \beta).$$

For $\hbar\omega \ll m_e c^2/\gamma$ (Thomson regime), $\omega^{**} \approx \omega^*$ and

$$\omega'_{\max} \approx \gamma^2(1 + \beta)^2 \omega \approx 4\gamma^2 \omega.$$

$$\boxed{\hbar\omega'_{\max} \approx 4\gamma^2 \hbar\omega}$$

Numerical: $W = 2 \text{ GeV}$, $\gamma = W/(m_e c^2) = 2000/0.511 = 3914$. $\lambda = 1 \text{ cm}$:

$$\lambda'_{\min} = \lambda/(4\gamma^2) = \frac{1 \times 10^{-2} \text{ m}}{4 \cdot 3914^2}.$$

$$4\gamma^2 = 4 \cdot 1.532 \times 10^7 = 6.13 \times 10^7.$$

$$\lambda'_{\min} = \frac{10^{-2}}{6.13 \times 10^7} \text{ m} = 1.63 \times 10^{-10} \text{ m}.$$

$$\boxed{\lambda'_{\min} \approx 0.16 \text{ nm}}$$

Question 3 — Doppler, group/phase velocity, photon emission, EM wave in medium

(a) Three SR tests in astrophysics; relativistic Doppler

Three phenomena: (i) transverse Doppler shift in fast-moving stars/binary pulsars (frequency redshift at $\theta = \pi/2$, $\omega' = \omega/\gamma$); (ii) aberration of starlight (annual stellar aberration); (iii) gravitational/cosmological time dilation observed in supernova light-curve stretching at high z (or relativistic beaming in AGN jets).

Photon 4-vector $K^\mu = (\omega/c)(1, \hat{n})$. $K \cdot U$ is invariant; in S the photon makes angle θ with $\mathbf{v}$:

$$K \cdot U = \gamma\omega(1 - \beta \cos \theta).$$

In S' the source is at rest, $U'^\mu = (c, \mathbf{0})$, and $K \cdot U = \omega'$:

$$\omega' = \gamma\omega(1 - \beta \cos \theta).$$

$$\boxed{\omega' = \gamma\omega(1 - \beta \cos \theta)}$$

(b) Phase and group velocity

$\omega = \omega(k)$. Definitions:

$$v_{\text{ph}} = \omega/k, \quad v_{\text{gr}} = d\omega/dk.$$

Vacuum dispersion ($n = 1$): $\omega = ck$, so $v_{\text{ph}} = c$ and $v_{\text{gr}} = c$:

$$v_{\text{ph}} v_{\text{gr}} = c^2$$

(c) Electron–proton fusion with photon emission

Lab-frame initial state, electron and proton head-on along $\hat{z}$:

$$P_e = (E_e/c, 0, 0, p_z), \quad P_p = (E_p/c, 0, 0, -p_z).$$

Both have the same $|p_z|$ (opposite directions, given $\mathbf{v}_e = -\mathbf{v}_p$). With masses different, demanding equal $|v|$ gives different $|p|$ unless we set $|p_e| = |p_p| = p$.

The cleanest reading: electron and proton have the same speed v_z but opposite directions. Then $p_e = \gamma m_e v_z$, $p_p = \gamma M_p v_z$ with the *same* γ .

After collision: single particle of total 4-momentum P_X , plus photon $K = (\hbar\omega_\gamma/c)(1, 1, 0, 0)$ along $\hat{x}$ with $\omega_\gamma = 2\pi c/\lambda$.

Conservation:

$$P_e + P_p = P_X + K.$$

Take $|\cdot|^2$ on both sides; with $K \cdot K = 0$:

$$(P_e + P_p)^2 = P_X^2 + 2P_X \cdot K.$$

Initial invariant mass squared:

$$(P_e + P_p)^2 = m_e^2 c^2 + M_p^2 c^2 + 2P_e \cdot P_p.$$

$P_e \cdot P_p = E_e E_p / c^2 - \mathbf{p}_e \cdot \mathbf{p}_p = E_e E_p / c^2 + p_e p_p$ (anti-parallel), with $\gamma_e = \gamma_p = \gamma$:

$$2P_e \cdot P_p / c^2 = 2\gamma^2 (m_e M_p) (1 + \beta^2).$$

Combined invariant mass:

$$s = (E_{\text{tot}}/c)^2 - 0 = (E_e + E_p)^2 / c^2 \quad (\text{net 3-momentum is zero}).$$

Net momentum is zero only if $p_e = p_p$, i.e. $m_e = M_p$ (false). Use instead the proper CM total: net 3-momentum

$$\mathbf{P}_{\text{tot}} = (p_e - p_p)\hat{z} = \gamma v_z (m_e - M_p)\hat{z}.$$

$sc^2 = E_{\text{tot}}^2 - |\mathbf{P}_{\text{tot}}|^2 c^2$. Energy conservation gives $E_{\text{tot}} = E_X + \hbar\omega_\gamma$ and momentum conservation gives $\mathbf{P}_X = \mathbf{P}_{\text{tot}} - \hbar\omega_\gamma \hat{x}/c$.

For the *minimum* total initial energy: the final composite particle is at rest in the CM of the (electron+proton+photon) system. Photon emission at $\theta = \pi/2$ to the collision axis, single-particle final state, minimum E_{tot} corresponds to threshold where the composite carries no further KE in the appropriate frame.

Invariant manipulation: form $(P_e + P_p - K)^2 = P_X^2 = M_X^2 c^2$. The minimum corresponds to the lightest possible M_X , namely $M_X = m_e + M_p$ (rest mass sum, no internal excitation):

$$(P_e + P_p)^2 - 2K \cdot (P_e + P_p) = (m_e + M_p)^2 c^2.$$

Compute $K \cdot (P_e + P_p) = (\hbar\omega_\gamma/c)(E_{\text{tot}}/c) - \hbar\mathbf{k}_\gamma \cdot \mathbf{P}_{\text{tot}}$. With $\mathbf{k}_\gamma \perp \hat{z}$ and $\mathbf{P}_{\text{tot}} \parallel \hat{z}$:

$$K \cdot (P_e + P_p) = \hbar\omega_\gamma E_{\text{tot}}/c^2.$$

Take $(P_e + P_p)^2 c^2 = E_{\text{tot}}^2 - |\mathbf{P}_{\text{tot}}|^2 c^2$:

$$E_{\text{tot}}^2 - |\mathbf{P}_{\text{tot}}|^2 c^2 - 2\hbar\omega_\gamma E_{\text{tot}} = (m_e + M_p)^2 c^4.$$

For minimum: assume the composite is produced at rest (in lab frame), so $\mathbf{P}_{\text{tot}} = \hbar\mathbf{k}_\gamma \Rightarrow |\mathbf{P}_{\text{tot}}|c = \hbar\omega_\gamma$, but $\mathbf{P}_{\text{tot}} \parallel \hat{z}$ and $\mathbf{k}_\gamma \perp \hat{z}$ are perpendicular, contradicting. The minimum compatible with kinematics is $\mathbf{P}_{\text{tot}} = 0$, achievable only if $m_e\beta\gamma = M_p\beta\gamma$, false. So $\mathbf{P}_{\text{tot}} \neq 0$.

Threshold (bare minimum): take electron and proton at the same $|v_z|$ but adjust so the photon's transverse momentum balances against the longitudinal $\mathbf{P}_{\text{tot}}$. In fact transverse and longitudinal components decouple; momentum conservation perpendicular to $\hat{z}$ requires $\mathbf{P}_X \cdot \hat{x} = -\hbar\omega_\gamma/c$, while parallel: $P_{X,z} = (p_e - p_p)$. Minimising E_{tot} at fixed ω_γ :

$$E_{\text{tot,min}} = \hbar\omega_\gamma + \sqrt{(m_e + M_p)^2 c^4 + (\hbar\omega_\gamma)^2}$$

(threshold: composite emerges with momentum exactly cancelling the photon, mass $m_e + M_p$, in the CM where $\mathbf{P}_{\text{tot,initial}} = \mathbf{0}$; the assumption $p_e = p_p$ is the choice that places us in this CM and is implicit in "minimum").

$$E_{\text{tot,min}} = \hbar\omega_\gamma + \sqrt{(m_e + M_p)^2 c^4 + (\hbar\omega_\gamma)^2}, \quad \omega_\gamma = 2\pi c/\lambda$$

(d) Potentials and 4-vector for plane wave in medium

Wave: $E_y = E_0 \cos(\omega t - kz)$, $B_x = B_0 \cos(\omega t - kz)$, with $|\mathbf{E}_0 \times \mathbf{B}_0| = 1$. Poynting along $+\hat{z}$: $\hat{E} \times \hat{B} = \hat{y} \times \hat{x} = -\hat{z}$. To have power flow along $+\hat{z}$ we take $B_x < 0$, so $\mathbf{E} \times \mathbf{B} = E_0 |B_0| \hat{z}$.

In a medium, $|\mathbf{B}| = n|\mathbf{E}|/c$, so $E_0 \cdot B_0 = nE_0^2/c = 1$, hence $E_0 = \sqrt{c/n}$, $|B_0| = \sqrt{n/c}$ (in SI).

Lorentz gauge potentials, $\mathbf{E} = -\partial_t \mathbf{A} - \nabla\varphi$, $\mathbf{B} = \nabla \times \mathbf{A}$. Try $\varphi = 0$ and $\mathbf{A} = A_0 \hat{y} \sin(\omega t - kz)$:

$$\mathbf{E} = -\partial_t \mathbf{A} = -A_0 \omega \hat{y} \cos(\omega t - kz).$$

Match to $E_y = E_0 \cos(\omega t - kz)$:

$$A_0 = -E_0/\omega.$$

Check $\mathbf{B}$:

$$\mathbf{B} = \nabla \times \mathbf{A} = \partial_z A_y \hat{x} \times \hat{y} \cdot (\dots) = -\partial_z A_y \hat{x} = A_0 k \cos(\omega t - kz) \hat{x}.$$

With $A_0 = -E_0/\omega$, $B_x = -(E_0 k/\omega) \cos(\omega t - kz)$, magnitude $E_0 k/\omega = E_0/v_{\text{ph}} = nE_0/c = B_0$. Lorentz gauge $\partial^\mu A_\mu = \partial_t \varphi/c^2 + \nabla \cdot \mathbf{A} = 0$ holds since $\varphi = 0$ and $\mathbf{A} = A_y(z, t) \hat{y}$.

$$\varphi = 0, \quad \mathbf{A} = -\frac{E_0}{\omega} \sin(\omega t - kz) \hat{y}$$

Wave 4-vector ($k = n\omega/c$):

$$K^\mu = (\omega/c, 0, 0, n\omega/c)$$

$K \cdot K = (\omega/c)^2(1 - n^2) < 0$ for $n > 1$: space-like.

Frame in which $\mathbf{E} = 0$ or $\mathbf{B} = 0$: invariants $\mathbf{E} \cdot \mathbf{B} = 0$ (already, $\mathbf{E} \perp \mathbf{B}$) and

$$\mathbf{E}^2 - c^2 \mathbf{B}^2 = E_0^2 - c^2 B_0^2 = E_0^2(1 - n^2) < 0.$$

Since $\mathbf{E}^2 - c^2 \mathbf{B}^2 < 0$, in some frame $\mathbf{E} = 0$ (purely magnetic). Boost along $\hat{z}$ with $\beta = E_y/(cB_x) \cdot (\text{sign})$.
Field transformation: $E'_y = \gamma(E_y - vB_x)$ (for boost along $+\hat{z}$ with $\mathbf{v} = v\hat{z}$, recalling $\mathbf{B}$ along $\hat{x}$):

$$E'_y = 0 \Rightarrow v = E_y/B_x = E_0/B_0 = c/n.$$

$$\boxed{\mathbf{v} = (c/n)\hat{z}}$$

This is the phase velocity of the wave; not larger than c for $n \geq 1$, so the frame exists.

Question 4 — Field tensor, Pauli–Lubanski, Cherenkov, parallel charged wires

(a) $F^{\alpha\beta}$, Bianchi, Maxwell, Lorentz force

$A^\mu = (\varphi/c, \mathbf{A})$ with metric $(+, -, -, -)$. Fields:

$$\mathbf{E} = -\nabla\varphi - \partial_t\mathbf{A}, \quad \mathbf{B} = \nabla \times \mathbf{A}.$$

Field tensor $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$:

$$F^{0i} = E_i/c, \quad F^{ij} = -\epsilon^{ijk} B_k.$$

Cyclic identity: substitute the definition,

$$\partial^c F^{ab} + \partial^a F^{bc} + \partial^b F^{ca} = \partial^c \partial^a A^b - \partial^c \partial^b A^a + \partial^a \partial^b A^c - \partial^a \partial^c A^b + \partial^b \partial^c A^a - \partial^b \partial^a A^c.$$

Pair partials commute:

$$= 0.$$

This is the homogeneous pair: $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$.

4-current $J^\nu = (c\rho, \mathbf{j})$. Inhomogeneous Maxwell:

$$\partial^\mu F_{\mu\nu} = \mu_0 J_\nu.$$

(Or J_ν in natural units, as in the question's normalisation.) Components: $\nu = 0$ gives $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$; $\nu = i$ gives $\nabla \times \mathbf{B} - \mu_0 \epsilon_0 \partial_t \mathbf{E} = \mu_0 \mathbf{j}$.

Lorentz force per unit volume:

$$\mathbf{f} = \rho \mathbf{E} + \mathbf{j} \times \mathbf{B}.$$

Identify $f_\mu = J^\nu F_{\nu\mu}$ component-wise:

$$f_i = J^0 F_{0i} + J^j F_{ji} = c\rho \cdot (E_i/c) + j^j \cdot (-\epsilon_{jik} B_k) = \rho E_i + (\mathbf{j} \times \mathbf{B})_i.$$

$$\boxed{\mathbf{f} = \rho \mathbf{E} + \mathbf{j} \times \mathbf{B}}$$

Time component:

$$f_0 = J^j F_{j0} = j^j (-E_j/c) = -\mathbf{j} \cdot \mathbf{E}/c.$$

Physical meaning: $-f_0 \cdot c = \mathbf{j} \cdot \mathbf{E}$ is the rate of energy density transfer from field to charges (Joule heating per unit volume).

(b) $W^\mu P_\mu = 0$

$W^\mu = (\mathbf{s} \cdot \mathbf{p}, E\mathbf{s}/c)$. In the rest frame: $\mathbf{p} = 0$, $E = m_0 c^2$:

$$W_{\text{rest}}^\mu = (0, m_0 c \mathbf{s}), \quad P_{\text{rest}}^\mu = (m_0 c, \mathbf{0}).$$

Scalar product:

$$W \cdot P|_{\text{rest}} = 0 \cdot m_0 c - m_0 c \mathbf{s} \cdot \mathbf{0} = 0.$$

$W \cdot P$ is a Lorentz invariant. Hence in every frame:

$$\boxed{U_W^\mu P_\mu = W^\mu P_\mu / (m_0 c |\mathbf{s}|) = 0}$$

(c) **Cherenkov emission for $\gamma = 100$**

Single-photon emission by free electron requires $v_{\text{electron}} > v_{\text{phase}} = c/n$.

(a) $n = 1$: $v_{\text{phase}} = c$, electron has $v < c$, no real-photon emission allowed (4-momentum conservation forbids it for a free electron). *No emission.*

(b) $n = 2$: threshold $\beta > 1/n = 0.5$. Electron with $\gamma = 100$ has $\beta \approx 1$:

$$\beta = 0.99995 > 0.5.$$

Cherenkov emission occurs. Cone angle:

$$\cos \theta_C = \frac{1}{n\beta}.$$

Numerically:

$$\cos \theta_C = \frac{1}{2 \cdot 0.99995} \approx 0.50003.$$

$$\boxed{\theta_C = \arccos(1/(n\beta)) \approx 60^\circ}$$

(d) **Parallel charged wires**

(i) Co-moving frame S' : wire is stationary, charge density ρ' . By length contraction the lab-frame charge density satisfies $\rho = \gamma\rho'$ (charge invariant; length contracts), so $\rho' = \rho/\gamma$. Current $\mathbf{j}' = 0$:

$$\mathbf{E}'(r) = \frac{\rho'_\ell}{2\pi\epsilon_0 r} \hat{r} = \frac{\rho_\ell/\gamma}{2\pi\epsilon_0 r} \hat{r}, \quad \mathbf{B}' = 0.$$

(ρ_ℓ is line-charge density.)

$$\boxed{\mathbf{E}' = \frac{\rho_\ell}{2\pi\epsilon_0 \gamma r} \hat{r}, \quad \mathbf{B}' = 0}$$

(ii) Lab frame S : line charge ρ_ℓ moves with $\mathbf{v} = v\hat{z}$, current $I = \rho_\ell v$:

$$\mathbf{E} = \frac{\rho_\ell}{2\pi\epsilon_0 r} \hat{r}, \quad \mathbf{B} = \frac{\mu_0 I}{2\pi r} \hat{\phi} = \frac{\mu_0 \rho_\ell v}{2\pi r} \hat{\phi}.$$

Check via boost of S' fields with $\rho_\ell^{\text{lab}} = \gamma\rho'_\ell$ and $E_\perp = \gamma E'_\perp$, $B_\perp = \gamma \mathbf{v} \times \mathbf{E}'/c^2$ — gives the same.

$$\boxed{\mathbf{E} = \frac{\rho_\ell}{2\pi\epsilon_0 r} \hat{r}, \quad \mathbf{B} = \frac{\mu_0 \rho_\ell v}{2\pi r} \hat{\phi}}$$

Two identical parallel wires: in the co-moving frame both wires are at rest, only electric (Coulomb) repulsion. Boost to any other inertial frame: $\mathbf{E}^2 - c^2\mathbf{B}^2$ is invariant. In S' :

$$(\mathbf{E}^2 - c^2\mathbf{B}^2)|_{S'} = (E')^2 > 0.$$

This is an invariant; therefore in *no* frame can $\mathbf{E}$ vanish while $\mathbf{B}$ remains finite. Equivalently:

No frame exists with purely magnetic interaction.

The proof: $|\mathbf{E}|^2 - c^2|\mathbf{B}|^2$ and $\mathbf{E} \cdot \mathbf{B}$ are the two field invariants; in S' they are $(E')^2 > 0$ and 0 respectively; "purely magnetic" requires the first invariant to be ≤ 0 — contradiction.